

ScenThread: Weaving Smell into Textiles

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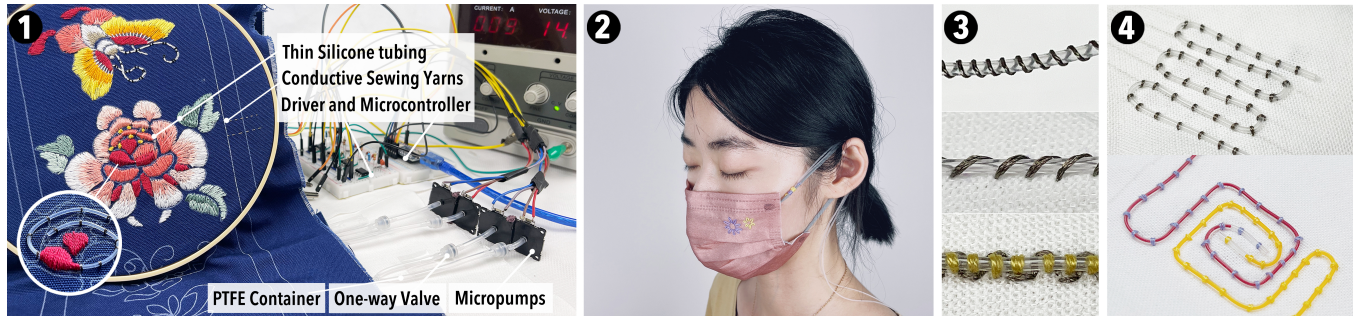


Figure 1: (1) The system design of *ScenThread*. It can be embedded into traditional embroidery work (Love of Butterfly) invisibly; (2) or merged into a mask releasing the scheduled scents for Olfaction Recovery Training in Post-COVID-19. (3) We explored the design patterns of the heating module, (4) intensity design (top), multi-scent, and mixed-scent releasing (bottom).

ABSTRACT

In this paper, we propose *ScenThread*, a controllable threadlike olfactory display that could be woven into textiles, creating a localized scent release on a flexible surface. *ScenThread* comprises a tubular PTFE scent reservoir, a lightweight and non-vibrating piezoelectric pumping system for delivering the scents, and a permeable silicone tubing to diffuse the aroma, as well as a heating module using conductive yarns for accelerating the scent release. *ScenThread* can be refilled and reused multiple times. We articulated the mechanical design of *ScenThread* and the pattern design of the *heating module*, *smell intensity* and *multi-scent release*. We conducted a preliminary study by testing the scent releasing performance through *maximum smell distance* and the *duration time*. We envision that *ScenThread* could integrate into a wide range of textile-based items in daily life naturally, seamlessly, and artistically.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

threadlike olfactory display; aromatic textiles; controllable scent releasing; permeable silicone tubing; piezoelectric pumping system; heating module; weaving techniques

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1 INTRODUCTION

In Human-Computer Interaction (HCI), the sense of smell as a powerful and promising medium has been explored and designed for immersive gaming[12], driving[5], well-being[2], notifications[7], sleep[1], and augmented flavor[10]. Researchers and designers developed various scent-delivery devices merging into different scenarios to provide a natural and seamless smell experience. The form factors vary from stabilized[6, 9] and portable devices[4] to small, lightweight and aesthetic wearables worn on the clothes[1], neck[2, 7], wrist[3], and even face[13]. Besides, textile is one of the most commonly used mediums in our everyday life, where we could always find on the garments, bed linings, furniture, shoes, jewelry, and vehicles. Perfume and fragrances are often added to daily textiles (e.g., aroma-enhanced fabrics in Textile Industry[14]) for decoration, cleaning, and refreshing. Some other researchers and artists have tried to engage smell into spatial textile interaction[8] and experimentally attach the tubes with scented water on the clothing[11]. We believe that there is a vast design space to integrate smell-enhanced technology into textiles within the HCI community. However, existing scent-delivery methods are still limited to their rigid form factor, the textile-based smell experience also needs further exploration.

Therefore, as a start of this promising area, we propose a novel threadlike olfactory display diffusing the aroma based on the permeability of the silicone tubing, aiming to create a localized scent release on a textile-based surface.

Table 1: Through test, we proved that the silicone tubing had a good performance for scent release with heat, even with a little amount of liquid aroma. A large smell distance we tested maybe caused by the pure essential oils we used.

Sample/ Aroma length inside the tube	Liquid-pumping Time	Maximum Smell Distance	Duration Time
A: 0.01ml liquid aroma with heat/ 1.57cm	0.86s	9.7cm	≈ 10min
B: 0.05ml liquid aroma with heat/ 7.85cm	4.2s	37cm	≈ 16min
C: 0.1ml liquid aroma with heat/ 15.72cm	8.4s	40cm	≈ 31min
D: 0.1ml liquid aroma without heat/ 15.72cm	8.4s	19.5cm	≈ 8h

2 DESIGNING SCENTHREAD

Lightweight and Non-vibrating Scent Delivery System. We built a lightweight and non-vibrating scent delivery system using the Piezoelectric Ultrasonic Micropump (Murata, MZB1001T02, 2cm × 2cm), driven by a NE555 Clock Chip converting a DC (14V) to an AC signal (14Vpp, 26khz), and controlled by a microcontroller (Arduino Nano) connecting to a laptop. We designed two piezo pumps as one unit, one is connected with the container for delivering the scented liquid with a certain amount (liquid-pumping); the other is to pump the air and push the liquid aroma to the target location (air-pumping). Multiple units can be designed for a multi-scent release.

Tubular Scent Diffuser and Container. Silicone is generally stable and non-reactive to an extensive range of chemicals and has a working temperature from -55 to 300°C. Besides, based on the micro-porous structure, silicone also has an ideal performance of permeability [15]. Therefore, we introduce the thin, flexible and accessible silicone tubing for encapsulating the pumped liquid aroma and releasing smell as a diffuser. Additionally, we choose Polytetrafluoroethylene (PTFE) tubing as the scent container for its stability and non-permeability, which has been studied and explored in previous work [13]. In our case, the tubular PTFE container, air-delivery tube, and the silicone tube are connected together through a Y-type three-way connector. We use two one-way valves for each unit, one is connecting the PTFE container and the liquid-delivery pump, and the other is placed between the air-delivery tubing and pump, avoiding unwanted movement of the liquid aroma inside.

Heating Module for Accelerating the Scent Release. We use silver fiber conductive sewing yarns (5Ω/cm) as the primary material of the heating module. To maximize the heating and scent releasing performance, the conductive yarns and the silicone tubing should be fully contacted.

3 SCENT RELEASING TEST

This test was conducted in an open office space with a room temperature of 26.3°C and humidity of 52%rh. We set up the experiment using a transparent food-grade silicone tubing (OUPLI, 24213, ID=0.9mm/OD=1.1mm). The scented liquid we used is the pure lavender essential oil (Plant Therapy) bought from Amazon. Four different samples (see Table1) were prepared and placed on four tables horizontally. We set all the samples with a 10s air-pumping time. We applied Sample A, B, and C with heat using the pattern of the top design from Figure 1-3, and set Sample D diffusing the scent naturally. The conductive yarn was connected to the micro-controller applying with a current of 0.12A, reaching a temperature

of 43°C. We recruited 6 participants from our lab (excluding those who had odor allergies), testing the samples in order and recording the smell distance (walking to each sample until had an obvious smell perception) every 5min of Sample A, B, C, and 15min for Sample D. We calculated the average value of the six maximum distance from the participants. The duration time was recorded after pumping the air until no liquid scent inside (equals to the heating time).

The results of the test (Table1) shows that *ScenThread* could provide both short term and long term scent release when applied with different amount of the scented material. Based on the smell intensity and distance, we could design the placement of *ScenThread* in order to shorten the smell-perceived time and involve *ScenThread* into both passive (with a long smell distance) and active (short smell distance) smell behavior. Moreover, We could also design a multi-scent and mixed-scent release guided by the maximum smell distance.

4 DESIGN EXPLORATIONS, APPLICATIONS, AND CONCLUSION

We conducted a small-scale design exploration by integrating *ScenThread* with conventional textile techniques. For the design of heating module, we could wrap the conductive yarns spirally along the outer side of the silicone tube (Figure 1-3, top); we could also embroider the tubing on the fabric by sewing the conductive yarns along (Figure 1-3, middle) or underneath the tubes (Figure 1-3, bottom), using conductive yarns as the fixing thread. Besides, *ScenThread* can release a single smell, multiple smells, and even mixing scents (Figure 1-4, bottom). We could also modify the scent intensity by using pure or diluted aroma, or controlling the amount of pumped scented materials, or choosing different design patterns of the tubes, such as one-thread or space-filling pattern (Figure 1-4, top). All above designs can be utilized for a second creation by combining with the decorative sewing thread.

We verified the feasibility of *ScenThread* with a preliminary test and a small-scale design exploration. We envision *ScenThread* could be coupled with a wide range of textile-based objects in our day-to-day life to provide interactive and localized scent releasing on the flexible surface with multiple odors (Figure 1-1,2). For example, aromatherapeutic pillow for sleep management, the scented curtain that could release dynamic and ambient smells, and aesthetic and personal jewelry and accessories.

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